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Section 1: TOPIC/MOTIVATION

- Starting point: Poverty persists not only because of lack of income, but because of constraints that prevent households from converting resources into welfare gains.
- Policy area/problem: Evaluating cash transfer (UCT) vs seed subsidy as poverty interventions in a developing country context (preliminary thoughts: Ethiopia/Nigeria)

<u>Why does this matter</u>:
Alleviating poverty is one of the core challenges of our times and the focus of development policy work. Within poverty reduction, there seems to be a constant debate on the subject of cash transfers:

a. Are cash transfers strictly superior to equivalent value resource gifts?
b. Is the flexibility of a cash transfer enough to make it the clear best choice for poverty intervention and alleviation?
c. What if certain resources/information, that are hard for the studied population to obtain, are worth more than the cost of their distribution?
d. What if we do a mix of cash transfers and resource distribution?

Other studies that address this overarching theme have been conducted, showing that is something that very much matters, and that is the motivation for our study. More new evidence is still needed because there isn't consensus one way or the other. The literature lacks clear experimental evidence on whether cash alone is sufficient to induce agricultural transformation, or whether complementary productive support is necessary. In particular, it remains unclear whether liquidity and input access constraints operate independently or interact multiplicatively. Studying this will allow us to understand better how future poverty interventions should be designed. In what contexts do we prefer UCTs and in what contexts are UCTs less efficient?

Section 2: RESEARCH QUESTION & DESIGN

When poor households receive resources (whether it be cash or product subsidy or both), what complementary frictions allow one type of resource distribution to impact poverty reduction better than the other?

Treatment/Exposure Variables (X variables):

- X1 - indicator variable indicating whether household received cash transfer (treatment group 1)
- X2 - indicator variable indicating whether household received seed subsidy + technological information on how to boost productivity (treatment group 2)

    - If both indicator variables have value of 1, then they are in treatment group 3
    - If both indicator variables have value of 0, then they are in control group
- Y (dependent variable) - monthly consumption expenditure which is being used as a proxy variable for welfare/income/poverty.

Since socioeconomic status/poverty is not easy to measure in general, and income is a volatile/difficult to observe measure of welfare, we shall use consumption data instead. Self reported by study participants once a month. The difference in consumption levels over a 12 month period (duration of intervention) will be the focus of the study and allow us to evaluate how our intervention(s) impacted poverty within the treatment group(s).

Other X variables (factors we want to control for):

- Distance of house to closest shopping market
- Size of farmland that they own/work on
- Indicator variable for whether they already have internet access
- Number of hours without electricity in household per month

Section 3: THEORY OF CHANGE (TC)

A. Market Failure & Complementary Frictions

The optimal adoption of high-yield technology is hindered by three interlocking barriers:

- Liquidity & Credit Constraints: Extreme poverty and lack of formal credit create chronic cash shortages, preventing farmers from affording the upfront costs of improved seeds during planting seasons.
- Information Gaps & Risk Aversion: Farmers face uncertainty regarding the returns of new seeds on their specific plots. Coupled with high income volatility from climate shocks, this forces households to stick with low-risk, low-yield traditional varieties.
- Behavioral Constraints: Immediate consumption and emergency needs often take precedence over long-term investment. This justifies a Cash-plus approach, where cash relaxes liquidity constraints while subsidies ensure resources are directed toward productive inputs.

B. The Causal Chain (Logic Model)

1. Intervention → Outcomes Path

- Intervention: Cash Transfer (UCT) + Seed Subsidy + Training.
- Mechanism: Cash provides immediate liquidity, while subsidies and training lower the risk and cost of adopting improved seeds, resulting in higher agricultural productivity.

- Impact: Surplus crops are sold in markets, leading to increased household consumption and improved long-term child nutrition.

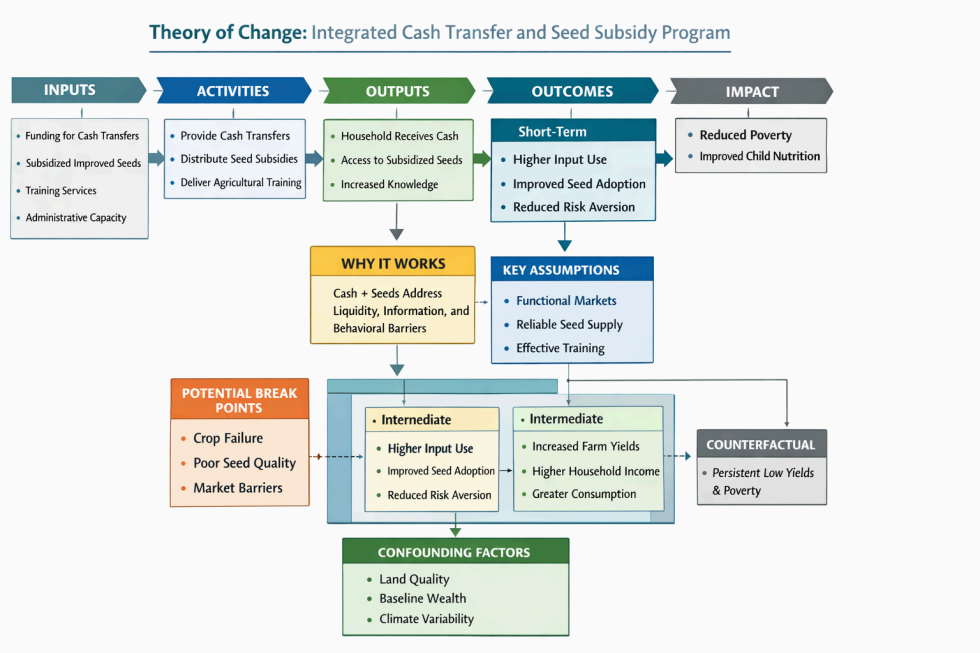
2. Key Assumptions

- Market & Price Stability: We assume functional local markets can provide complementary inputs (fertilizer/labor) and absorb surplus harvest without triggering significant inflation or a collapse in crop prices.
- Infrastructure & Information: We assume that extension training effectively resolves uncertainty regarding technology returns, and that infrastructure allows for market access without prohibitive transaction costs.
- Climate Resilience: We assume the intervention (e.g., stress-tolerant seeds) provides sufficient resilience to ensure that localized weather fluctuations do not entirely offset productivity gains.

3. Counterfactual & Confounders:

- Counterfactual: Without intervention, households remain trapped in subsistence farming with persistent malnutrition.
- Confounders: We recognize that pre-existing differences in land quality or household education could confound results. Our experimental design (such as randomization or quasi-experimental methods) will aim to isolate the program's impact from these baseline characteristics.

C. ToC Diagram



Section 4: LITERATURE BRIEF

Government of Ethiopia's public works and complementary programmes: A mixed-methods study on pathways to reduce intimate partner violence - ScienceDirect

- This is in the context of Intimate Partner Violence, so we can extrapolate its design to our study. The authors demonstrate that public works and complementary programming reduce poverty-related stress and influence intra-household dynamics, their focus is on IPV and social outcomes rather than productive investment. Moreover, the bundled nature of complementary programming makes it difficult to isolate which constraints are binding.
- Our project contributes by experimentally disentangling liquidity and agricultural input constraints to identify whether relaxing multiple frictions is necessary to generate sustained productivity gains among smallholder households

Integrated Nutrition and Social Cash Transfer (IN-SCT) Pilot in Ethiopia: perceptions and feedback from client

- This pilot demonstrates that combining cash transfers with nutrition-sensitive and behavioral components improves knowledge and service uptake, but structural constraints such as low productivity and limited market access continue to bind. Moreover, the bundled design does not isolate which constraints are most limiting for sustained poverty reduction. In our study, if UCT effects are small alone but large when combined with seed subsidy, that implies multiple binding constraints.

The Impact of Cash Transfers: A Review of the Evidence from Low- and Middle-income Countries | Journal of Social Policy

The study finds that cash transfers generally improve outcomes in the intended direction across most domains. They reduce monetary poverty, improve school attendance and health indicators, and in many cases increase savings and productive investment. It also finds little evidence that transfers reduce adult labor supply or create major unintended distortions. However, the evidence on sustained productivity gains and the mechanisms through which cash affects production remains mixed. In particular, it remains unclear whether liquidity alone is sufficient to induce adoption of high-return agricultural technologies in contexts where multiple frictions bind.

Other References:

The following literature is useful in the context of why pure cash subsidies are more productive when supplemented with other in-kind interventions as well as define some of the agricultural context of Sub-Saharan African countries' agricultural landscape.

What 'Cash Plus' Programs Teach Us About Fighting Extreme Poverty

Cash transfers: what does the evidence say?

Fighting urban poverty: Public works increase the welfare of the poor in Addis Ababa by 20% | VoxDev

Seeding Growth: Unlocking the Potential of Agricultural Commodity Exchanges in Sub-Saharan Africa